ORF 245: Fundamentals of Statistics – Spring 2025

Precept 4

1. (**Joint Distribution**)

   (a) Joint PMF $p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y)$ and Joint PDF $f_{X,Y}(x, y)$ (function satisfying $\mathbb{P}((X, Y) \in A) = \int_A f_{X,Y}(x, y)dxdy$.)

   (b) Marginal PMF $p_X(x) = \sum_{y \in \text{ Range } Y} p(x, y)$ and Marginal PDF $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)dy$

   (c) Joint can determine Marginal counterpart while the converse is false

   - $X \sim \text{Ber}(1/2)$, $Y_1 \sim \text{Ber}(1/2)$ indep of X, $Y_2 = X$: marginal same, joint different
   - The converse if true if they are independent.

   (d) Conditional PMF/PDF

   (e) Independence: X and Y are independent if $p_{X,Y}(x, y) = p_X(x)p_Y(y)$ (for discrete R.V.) and $f_{X,Y}(x, y) = f_X(x)f_Y(y)$ (for continuous R.V.)

   - That also holds for CDF.

   (f) Expectation: $\mathbb{E}[h(X, Y)] = \sum_{x,y} h(x, y)p_{X,Y}(x, y)$; $\mathbb{E}[h(X, Y)] = \int h(x, y)f_{X,Y}(x, y)dxdy$

   - Linearity of Expectation: $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$.
   - Covariance: $\mathbb{C}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$.
     - Independence $\to$ Covariance $= 0$ (uncorrelated), the converse if not true (but is true for joint Gaussian random variable)
     - $\mathbb{V}(X + Y) = \mathbb{V}(X) + \mathbb{V}(Y) + 2\mathbb{C}(X, Y)$: uncorrelated, further have $\mathbb{V}(X + Y) = \mathbb{V}(X) + \mathbb{V}(Y)$.

   (g) Correlation Coefficient: $\rho_{X,Y} = \frac{\text{Cov}(X,Y)}{\sigma_X \cdot \sigma_Y} = \frac{\mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]}{\sqrt{\mathbb{E}[(X - \mathbb{E}X)^2] \cdot \mathbb{E}[(Y - \mathbb{E}Y)^2]}}$
   $\rho_{XY} \in [-1, 1]$. A value of +1 implies perfect positive correlation (linear relationship), and a value of -1 implies perfect negative correlation (linear relationship). A value of 0 implies no correlation.

2. (**A uniform distribution over a square region**) Let (X, Y) be uniformly distributed in the region $\{(x, y)||x| + |y| < 1\}$.

   - What is the PDF of (X, Y)?
   - What is the covariance of X and Y?
   - What are the expectations of X and Y? What is the conditional PDF of X given $Y = 1/2$?

   - **Solution**:
     - PDF of (X, Y)
       $$f(x, y) = \begin{cases} 1/2 & \text{if } |x| + |y| < 1 \\ 0 & \text{otherwise} \end{cases}$$
     - $\text{cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0$
     - $\mathbb{E}X = \mathbb{E}Y = 0$; $f_X(x|Y = 1/2) = \begin{cases} 1 & \text{if } |x| < 1/2 \\ 0 & \text{otherwise} \end{cases}$

3. (**Linearity of Expectation**) Calculate the expected number of cards that need to be turned over in a regular 52-card deck to see the first ace.

- **Solution**:
  - X number of cards that need to be turned over in a regular 52-card deck to see the first ace.
  - We define $X_i = 1\{\text{card } i \text{ is turned over before first A}\}$, for example:

    $$X_1 = 1\{\text{2 of diamonds is drawn before first Ace}\}$$

    $$X_2 = 1\{\text{2 of hearts is drawn before first Ace}\}.$$

    And so forth.
    We then have $X = \sum_{i=1}^{48} X_i + 1$.
  - $\mathbb{E}[X] = 48\mathbb{E}[X_1] + 1$ by linearity and symmetry.
  - $\mathbb{E}[X_1] = \mathbb{P}(\text{2 of diamonds is drawn before first Ace}) = 1/5$.

4. (**Q&A**)